Report 5: The Sampled Receiver Against the Continuous Integral

EE451 — Zak Transform / Delay–Doppler Communication
Simulation S5

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1 What this report is about

- The slides first define the receiver as a continuous integral (a matched filter), then replace it by a sum over ADC samples with the symbol "≈".
- How good is that "≈"? How many samples per T_s (the oversampling factor D) are needed?
- This report answers it, and also explains a trap in the earlier simulation (Report 4) at $D = 1$.

2 The concept in brief

- **Ideal receiver.** For each delay bin m and period q, multiply the received waveform $r(t)$ by the conjugate pulse g^*, centred at $mT_s + qT$, and integrate over time. This gives $u_m[q]$.
- **Real receiver.** Hardware only has samples $r[a] = r(a\,T_{\text{samp}})$, with $T_{\text{samp}} = T_s/D$. The integral becomes a sum, $u_m[q] \approx T_{\text{samp}} \sum_a r[a]\, g^*[a - Dm - Pq]$.
- **It is a Riemann sum.** The sum is the integral approximated by rectangles of width T_{samp}. A finer grid (larger D) gives a better approximation.
- **When is it exact?** If the integrand varies smoothly enough (it is band-limited), sampling at twice the bandwidth is already exact. Here the bandwidth is about $1/T_s$, so $D \geq 2$ should be enough, and $D = 1$ should fail through aliasing.
- **Cutting the pulse.** A real pulse has finitely many taps. The cut creates a sharp edge, so the sum is no longer exact; its error should shrink as D grows.
- **The three slide forms are one computation.** Slide 4 (sum over all samples), slide 5 (FIR with L_g taps around each centre) and slide 6 (the constant T_{samp} absorbed into the pulse) only re-write the same sum.
- **The $D = 1$ trap.** The Gauss×sinc pulse is zero at every integer multiple of T_s. At $D = 1$ the sampled pulse is therefore a single non-zero sample, which looks perfectly orthogonal even though the real pulse is not.

One-line memory aid: the ADC-rate sum is a Riemann sum of the matched-filter integral. It is exact (to rounding) once $D \geq 2$ for a band-limited pulse, a cut pulse converges roughly as $1/D^2$, and $D = 1$ must not be trusted.

3 What was simulated

- Grid M=16, N=8; Gauss×sinc pulse with $\alpha = 4$, once untruncated and once cut at $\pm 6\,T_s$.

- A very fine sum ($D = 64$) stands in for the true integral and is the reference.
- For $D = 1, 2, 4, 8, 16$ the sampled projection $u_m[q]$ is compared with the reference, with the pulse aligned to the sample grid and with a $0.37\,T_s$ time offset.
- The S4 discrete model (Report 4) is evaluated at the same D values and compared with the continuous-time Nyquist residual ε.
- The three slide forms (slides 4–6) are checked to give identical output.

4 Results and discussion

4.1 What the sampled sum actually adds up

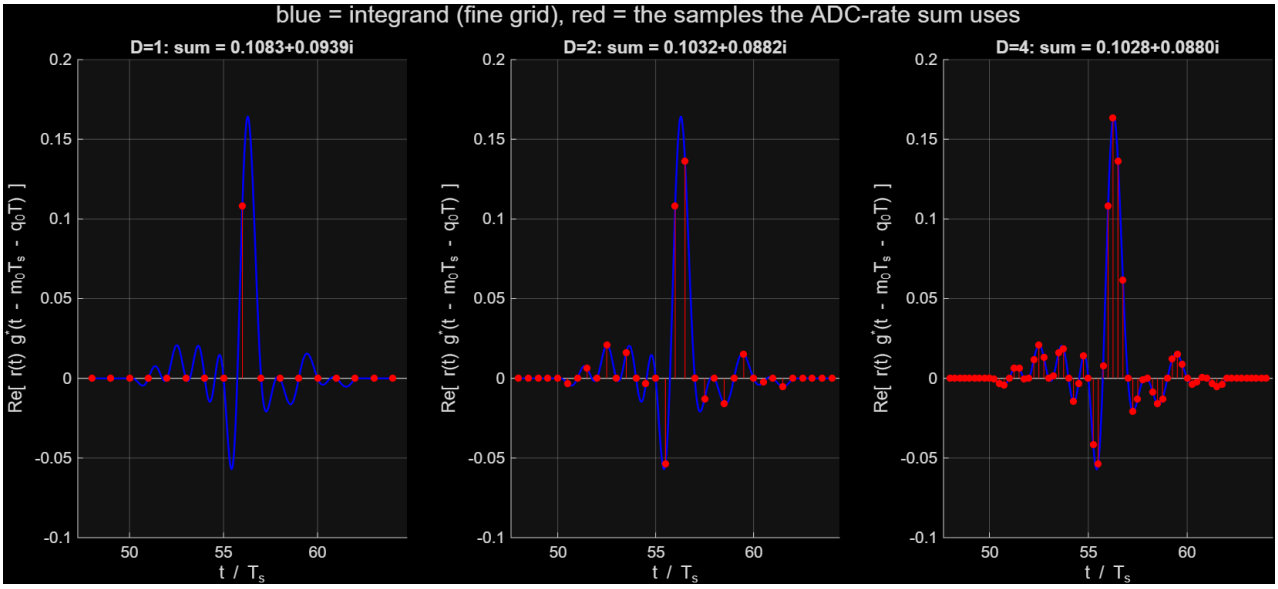


Figure 1: One projection $u_m[q]$. Blue: the integrand on a fine grid. Red: the samples that the ADC-rate sum uses, for $D = 1, 2, 4$. Cut pulse, $(m_0, q_0) = (8, 3)$.

- **Expected:** more samples per T_s means a better picture of the integrand, so the sum should settle to a fixed value.
- **Observed:** with $D = 1$ the red samples land at integer multiples of T_s and miss the peaks and dips of the blue curve; the sum is $0.1083 + 0.0939j$. With $D = 2$ and $D = 4$ it is $0.1032 + 0.0882j$ and $0.1028 + 0.0880j$, which are essentially the same. The $D = 1$ value differs from the $D = 4$ value by about 6 %.
- Takeaway: $D = 1$ misses detail of the integrand; $D \geq 2$ already captures it.

4.2 Error against D

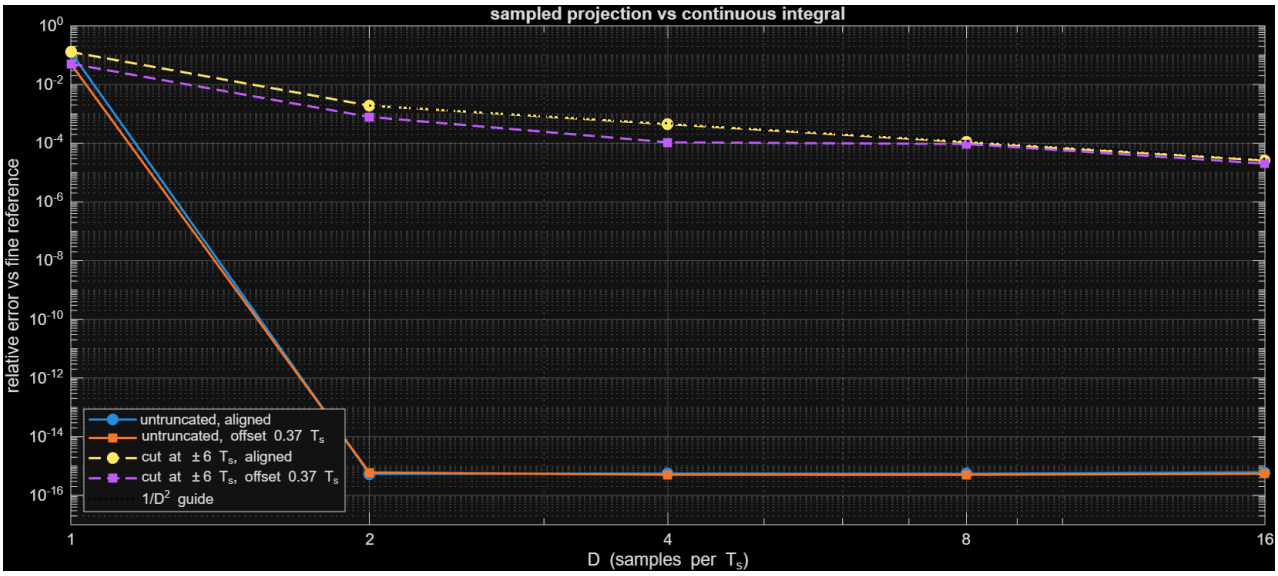


Figure 2: Relative error of the sampled projection against the fine reference, for the untruncated and the cut pulse, each aligned and offset by $0.37\,T_s$.

- **Expected:** the untruncated (band-limited) pulse is exact for $D \geq 2$; the cut pulse improves gradually.
- **Observed:** the untruncated curves drop from about 6–14 % at $D = 1$ straight to rounding level ($\sim 10^{-16}$) at $D = 2$ and stay there. The cut pulse falls from 1.9×10^{-3} at $D = 2$ to 2×10^{-5} at $D = 16$ (aligned case), roughly as $1/D^2$ (slightly faster here).
- The result is the same whether the waveform is aligned with the sample grid or offset by $0.37\,T_s$.
- Takeaway: $D \geq 2$ is the real threshold. Beyond it, any remaining error comes only from cutting the pulse, and is small.

4.3 Where the remaining error sits

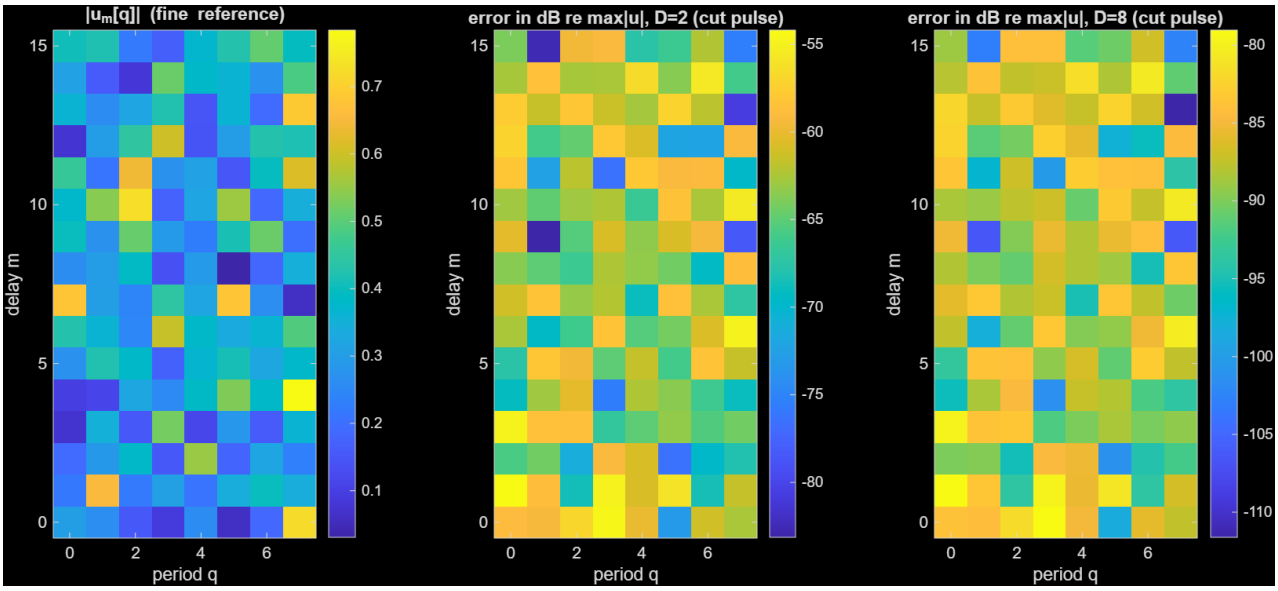


Figure 3: Left: $|u_m[q]|$ from the fine reference. Middle and right: error of the sampled projection (cut pulse) in dB relative to $\max|u|$, for $D = 2$ and $D = 8$.

- **Expected:** the cut-pulse error is small and shrinks with D.
- **Observed:** the error is about -55 to $-82\,\mathrm{dB}$ at $D = 2$ and -80 to $-112\,\mathrm{dB}$ at $D = 8$ (relative to the largest $|u|$), i.e. about $25\,\mathrm{dB}$ lower. It looks noise-like along the period q, with somewhat larger values in the lowest delay rows (we did not investigate why).
- Takeaway: even at $D = 2$ the sampled projection is accurate to better than $0.2\,\%$.

4.4 Does the earlier (D=1) result survive? The trap

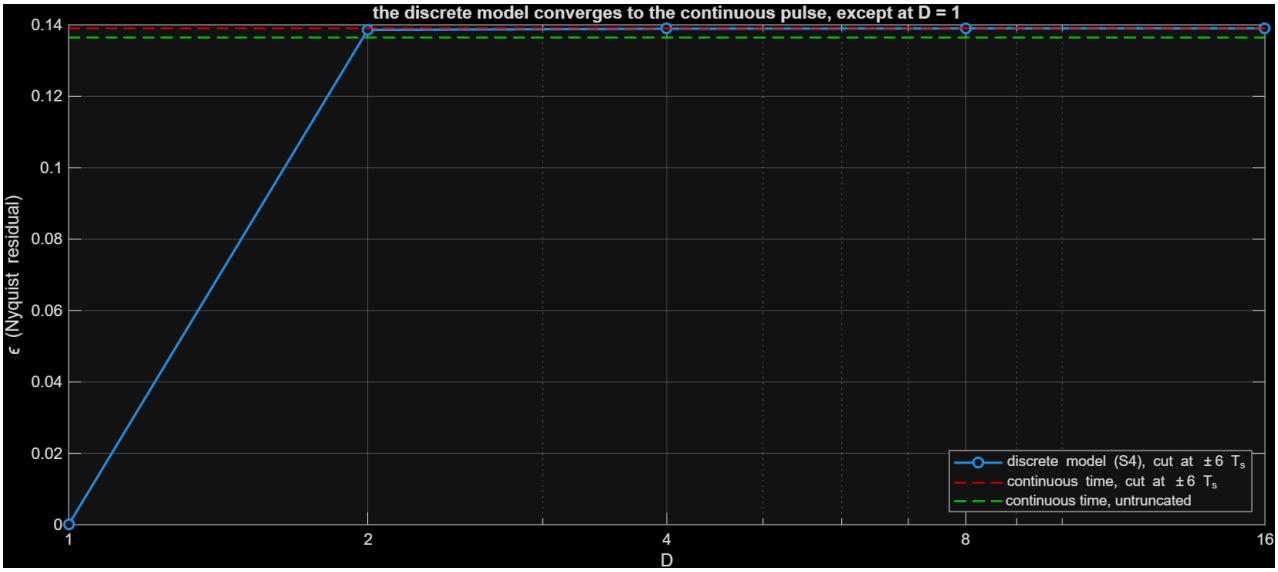


Figure 4: Nyquist residual ε of the Report 4 discrete model against D (blue), against the continuous-time value (dashed).

- **Expected:** the discrete model should agree with the continuous pulse at every D.
- **Observed:** for $D = 2, 4, 8, 16$ it gives 0.1386, 0.1389, 0.1390, 0.1390, converging to the continuous value 0.1390 for the cut pulse (0.1364 untruncated). At $D = 1$ it gives ≈ 0.
- **Why:** at $D = 1$ the sampled Gauss×sinc is 1 at the centre and 0 at all other samples, so the discrete model sees a perfect pulse. The true pulse is not orthogonal; the sampling just hid it (aliasing).
- The same effect hides the normalisation. With the unit energy $\int |g|^2 = 1$ of the continuous pulse, the $D = 1$ samples have energy $1/0.9437$, so the loopback returns $Y \approx 1.06\,X$, i.e. $\varepsilon = 1/0.9437 - 1 = 0.0596$. The script prints the measured $D = 1$ error ($\|Y - X\|/\|X\|$, untruncated pulse) next to this formula; our independent check gave 0.0596 for both.
- Takeaway: results at $D = 1$ must not be trusted for this pulse. The simulations S4 and S6–S8 use $D = 4$, where the discrete and continuous models agree.

4.5 Slides 4, 5 and 6 are the same computation

- **Expected:** the full sum over all samples (slide 4), the L_g-tap FIR (slide 5) and the version with T_{samp} inside the pulse (slide 6) differ only by bookkeeping.
- **Observed:** the script asserts that the three outputs agree to better than 10^{-10}; our independent check gave 1.7×10^{-15} (rounding level).

5 Summary

- The slides' sampled receiver is a Riemann sum of the continuous matched-filter integral; the "≈" is accurate.
- For a band-limited pulse it is exact to rounding already at $D = 2$; for a pulse cut at $\pm 6\,T_s$ the error is 2×10^{-3} at $D = 2$ and falls roughly as $1/D^2$.
- The slides' Gauss×sinc pulse is genuinely not orthogonal ($\varepsilon \approx 0.14$). At $D = 1$ the discrete model wrongly hides this ($\varepsilon \approx 0$), so $D \geq 2$ must be used.
- The three sampled-receiver forms in the slides agree to rounding level.

Appendix: details and assumptions

- Pulse: $g(t) = e^{-\pi\alpha(t/T)^2}\,\mathrm{sinc}(t/T_s)$ with $T = MT_s$, as assumed in Report 4 (the slides do not give the widths). Unit energy is imposed in continuous time, $\int |g|^2 dt = 1$; the discrete model of Report 4 instead normalises the sampled pulse, $\sum |g[\ell]|^2 = 1$, which at $D = 1$ forces the single unit sample.
- The reference is a sum at $D = 64$ (in an independent check, the untruncated reference at $D = 32$ and $D = 64$ agree to 2×10^{-16}, and the cut one to 5×10^{-6}), so the quoted errors are with respect to this reference, not an analytic integral.
- The band-limit argument is approximate: a Gaussian window slightly widens the sinc spectrum, but for $\alpha = 4$ the effect is negligible (the untruncated error is at rounding level for $D \geq 2$).
- The continuous-time ε values (0.1364 untruncated, 0.1390 cut) are $\sqrt{\sum_k |R(kT_s) - \delta_k|^2}$ from a fine-grid autocorrelation. In an independent check, the loopback $\|Y - X\|/\|X\|$ of one random frame at $D \geq 2$ was 0.131 (untruncated) and 0.134 (cut); this differs from the values above by a few percent because it is a single random realisation.
- The transmit waveform is built on a circular frame of length $MN\,T_s$, so pulses wrap around the frame edge; the integration window of Fig. 1 is $\pm 8\,T_s$ around the pulse centre.
- Report 4's D sweep used a half-span of $4\,T_s$ (values 0.153, 0.157, 0.157 at $D = 2, 4, 8$); Fig. 4 here uses $6\,T_s$ (hence 0.139). The behaviour at $D = 1$ is the same.
- The figures come from `S5_sampled_vs_continuous.m` (MATLAB dark theme).